

Question 1 (Structures – Declaration & Initialization)

Declare a structure named `Student` with members: `rollNo` (integer), `name` (string), and `cgpa` (float). Then define a structure variable and initialize it with `rollNo = 101`, `name = "Ali"`, `cgpa = 3.65`.

Question 2 (Nested Structures)

Define an inner structure `Date` containing `day`, `month`, `year` (integers). Then define an outer structure `Employee` containing `id` (int), `name` (string), and a nested `Date` object for the joining date. Write code to declare an `Employee` variable and assign joining date as 15, 6, 2021.

Question 3 (Enumerations)

Create an enumeration `Color` with values `RED`, `GREEN`, `BLUE`. Write a C++ code snippet that declares an enumeration variable `myColor`, assigns `GREEN` to it, and then uses a switch statement to print "Go" if `GREEN` is selected.

Question 4 (Classes – Defining & Using a Class)

Define a class `Rectangle` with private members: `length` and `width` (float). Provide public member functions:

- `setDimensions(float l, float w)` to set values.
- `float area()` to return the area.

Write the complete class definition and then show how to create an object of `Rectangle` and call both functions.

Question 5 (Member Functions Defined Outside the Class)

Rewrite the `Rectangle` class of Question 4 such that the member functions are declared inside the class but defined outside the class using the scope resolution operator (`::`).

Question 6 (Constructors & Overloaded Constructors)

Create a class `Book` with private members: `title` (string) and `price` (float). Provide:

- A default constructor that sets `title = "Unknown"` and `price = 0.0`.
- A parameterized constructor that accepts a title and a price.
- A member function `display()` to print the details.

Write the complete code and show how both constructors are invoked.

Question 7 (Default Copy Constructor – Theory)

Explain what the default copy constructor does in C++. Give a small example where the default copy constructor is automatically invoked when an object is passed by value to a function.

Question 8 (Destructors)

Write a class `Logger` that prints "Object created" inside the constructor and "Object destroyed" inside the destructor. Create a local object inside `main()` and observe the order of output. (Write the full code.)

Question 9 (Static Class Data)

Define a class `Counter` that has:

- A private static integer `count` that keeps track of how many objects of the class have been created.
- A public constructor that increments `count`.
- A static member function `getCount()` that returns the total object count.

Write the code and then show how to access `getCount()` without creating any object.

Question 10 (Inheritance – Base & Derived Class)

Create a base class `Vehicle` with a protected member `fuelCapacity` (float) and a public function `setFuel(float f)`.

Derive a class `Car` from `Vehicle` that adds a private member `numberOfDoors` (int). In `Car`, write a function `showDetails()` that displays fuel capacity and number of doors. Write the complete code and show how an object of `Car` accesses the base class member function `setFuel()`.